

Copernicus Global Land Operations

“Cryosphere and Water”

”CGLOPS-2”

Framework Service Contract N° 945954 (JRC)

PRODUCT USER MANUAL

LAKE WATERS

300M PRODUCT

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GCOS	Global Climate Observing System
GIO	GMES (Global Monitoring for Environment and Security) Initial Operations
GLWD	Global Lakes and Wetlands Database
GPT	Graph Processing Tool
LSR	Lake Surface Reflectance
LWQ	Lake Water Quality
MERIS	Medium Resolution Imaging Spectrometer
MSI	Multispectral Instrument
NERC	National Environment Research Council (UK)
NRT	Near Real Time
NTU	Nephelometric Turbidity Unit
obs	Observation
OC	Ocean Colour
OLCI	Ocean and Land Colour Instrument
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAA	Quasi-Analytical Approach
QAR	Quality Assessment Report
rep	representative
R_w	Water-leaving reflectance
TS	Trophic State
TSI	Trophic State Index

TSM	Total Suspended Matter
TUR	Turbidity
WGS84	World Geodetic System 1984

1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) Lake Water Quality 300m (LWQ-300m) product provides an optical characterisation of more than 4200 inland waterbodies. These waterbodies belong to the world's largest or are otherwise of specific environmental monitoring interest. The set of water bodies that are included in the observations consist for the largest part of natural lakes. However, reservoirs, enclosed lagoons and wide river sections are also included. The LWQ-300m product collection includes LWQ-300m v1.3 based on Envisat MERIS (May 2002-April 2012) and Sentinel-3 Ocean and Land Colour Instrument (OLCI) (April 2016-December 2019), superseded by S-3 OLCI in v1.4 (June 2020-October 2024), v2.0 (September 2024-October 2025) and the current v2.1.

Version 2 of the product contains six (sets of) variables: lake water-leaving reflectance (all wavebands describing water properties, after atmospheric correction), turbidity, total suspended matter concentration, chlorophyll-a concentration, a floating cyanobacteria index and a trophic state index (derived from phytoplankton biomass by proxy of chlorophyll-a). Some of these estimates now benefit from uncertainty models developed in the context of the ESA Lakes Climate Change Initiative (Lakes_cci), providing per-pixel estimates of relative uncertainty.

Production and delivery of the variables are set over 10-day intervals (starting the 1st, 11th and 21st day of each month) and mapped to a common global spatial grid at nominally 300m (~0.0022°) spatial resolution. The algorithms used are implemented in the *Calimnos* processing chain v2.1 and were tuned and validated against 13 predefined optical water types (OWT), initially in the UK NERC UK GloboLakes project, and later in the ESA Lakes_cci project. This Product User Manual (PUM) describes the product collection, including the latest developments in the current product version.

1.2 SCOPE AND OBJECTIVES

One of the main objectives of the Copernicus programme is to provide to the scientific community and other stakeholders, including policy makers, state-of-the-art information required for their respective applications. Prior to LWQ-300m product v2, the products turbidity (TUR) and Trophic State Index (TSI) were already in pre-operational mode, whereas the water-leaving reflectance was in demonstration mode due to suspected limitations which do not ultimately affect the derived turbidity and TSI. New variables added in v2 include Total Suspended Matter (TSM) concentration, chlorophyll-a concentration and a floating cyanobacteria index, and are in demonstration mode pending further review. The original products are provided for the full Envisat MERIS and Sentinel-3 OLCI (A/B) archives as versions 1.3 and 1.4. Version 2 (now updated to v2.1) is delivered near real-time (NRT) using OLCI. The objective of this document is to provide details on the products and their usage.

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3 summarizes the retrieval methodology
- Chapter 4 describes the technical properties of the product
- Chapter 5 summarizes the results of the quality assessment including current limitations

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service ‘CGLOPS’ Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service.
CGLOPS2_ATBD_LWQ300_v2.1.0_I1.00	Algorithm Theoretical Basis Document of the Lake Water Quality 300m product.
CGLOPS2_QAR_LWQ300_v2.1.0_I1.00	Report describing the results of the scientific quality assessment of the Lake Water Quality 300m product.

2 REVIEW OF USER REQUIREMENTS

According to the applicable document [AD3], the user requirements relevant for Lake Water Quality (LWQ) products are:

Definition:

The Lake Water Products are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3 , Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3 , floating cyanobacteria index and lake surface reflectance spectra (LSR, given in terms of fully normalised water reflectance R_w ; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. In addition, the mean of relative uncertainty of Chl-a and TSM in the observation period, and the relative uncertainty and unbiased uncertainty in LSR at reflectance peak are included. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of each LSR waveband may yield a physically inaccurate spectrum of reflectance and would, thus, not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to monitoring eutrophication. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The pixel size of the baseline dataset shall be provided at a resolution of 300m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

Geographical coverage:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data. The following output specifications are further optimised with respect to the requirements:

- pixel size at 300m: 0.25/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 300m: 161280 columns, 80640 lines

Continental windows

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

Accuracy requirements:

Lake Water Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and trophic state has an uncertainty threshold of 0.3 in log10 or 50% [AD3].

Temporal Definition

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally ten and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

3 ALGORITHM

3.1 OVERVIEW

The Global Land Component of the Copernicus Land Service (CGLOPS) Lake Water processing uses the *Calimnos* processing chain. *Calimnos* was initially developed in [UK NERC GloboLakes](#) (2012-2018) with input from the ESA Diversity-II project (2015) and has since been further developed with core collaboration between Plymouth Marine Laboratory (PML), University of Stirling, Brockmann Consult (BC) and HYGEOS in the Copernicus Land Monitoring Service (CLMS), H2020 TAPAS, H2020 EOMORES, and ESA Lakes_cci activities. The chain can be used to process archived Envisat MERIS, Sentinel-3 OLCI, Aqua-MODIS and Sentinel-2 MSI data for operational processing. This document refers to 300m products based on OLCI generated using *Calimnos* v2.1, with comparable performance achievable for the period covered with MERIS which is currently not provided for this product version.

Calimnos combines into a single processing chain: data discovery, spatial resampling and subsetting by target area (individual water bodies, image granule or any other region of interest), radiometric and atmospheric corrections, pixel identification (land/cloud/water/ice), optical water type classification, individual algorithms (per parameter and water type), weighted blending of results, optional reprojection and spatiotemporal aggregation (Figure 1). For the production of 300m products *Calimnos* is operated on the Copernicus Data Space Ecosystem (CDSE).

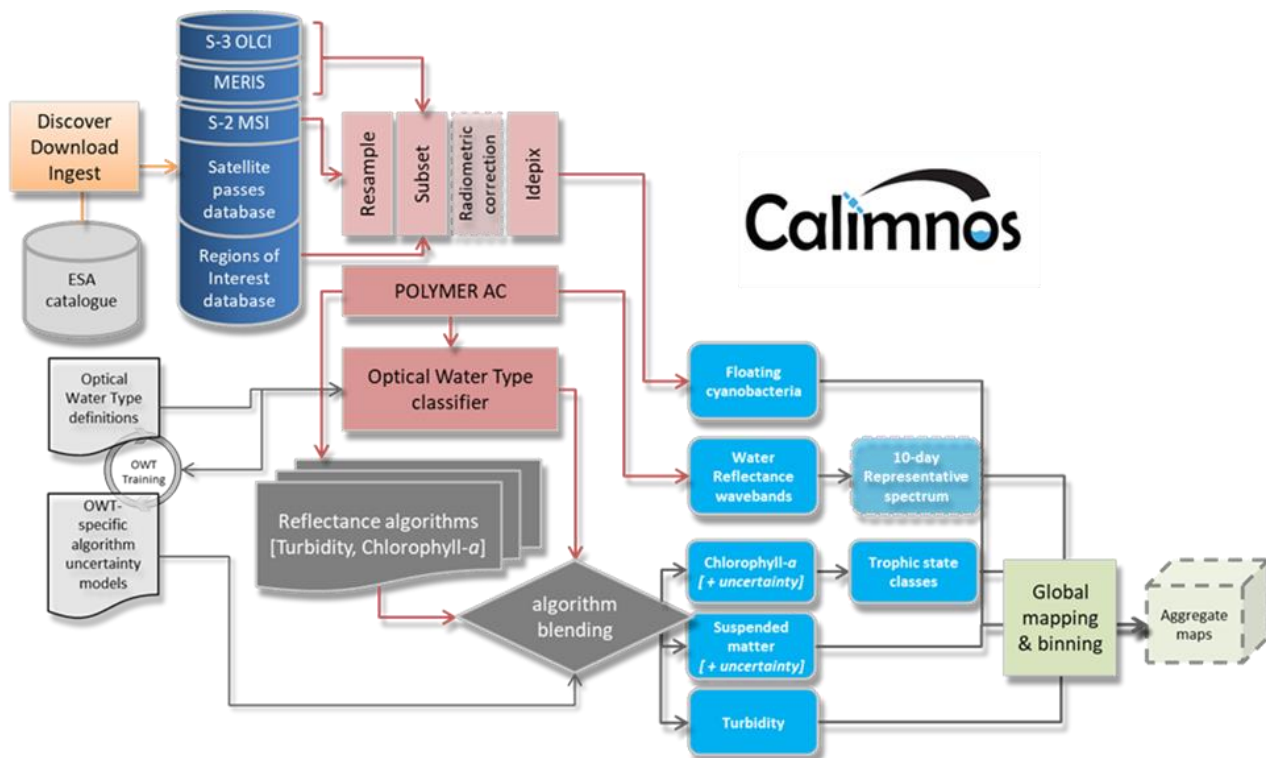


Figure 1: Schematic overview of the *Calimnos* processing chain v2.1 for 300m product generation

3.2 THE RETRIEVAL METHODOLOGY

3.2.1 Processing Chain and applied Tools

A schematic overview of *Calimnos* is given in Figure 1. The main processing stages and related software / processors / algorithms include:

- Data discovery.
 - o Newly downloaded L1b satellite passes are entered into a geospatial database. Target regions are similarly specified in a geospatial database and satellite products which overlap any of the target regions are queued for processing. In the context of re-processing, any duplicate passes are removed. The procedure relies on in-house Python scripts and PostgreSQL database functionality.
- Subsetting.
 - o Tool: SNAP Subset operator
 - o For best processing performance, satellite passes are subset to bounding boxes around each target area. The target area is defined as the maximum water extent observed in [ESA CCI Land Cover](#) product v4. The subsetting routine is part of the SNAP toolbox, called through the Graph Processing Tool (GPT).
- Radiometric corrections.
 - o Tools:
 - MERIS L1b Radiometric Correction v5.0.3, comprising a smile correction following the MERIS ATBD and a radiometric equalisation following Bouvet and Ramoino (2010)
 - OLCI: Python (for applying vicarious calibration coefficients, separated between OLCI-A/B)
 - o Any radiometric corrections defined by ESA and/or EUMETSAT following the release of the data are applied to the L1b imagery before submitting the data to atmospheric correction. System-vicarious calibrations are implemented (identical to those used in the [ESA CCI Ocean Colour](#) project).
- Pixel identification.
 - o Tools:
 - MERIS: SNAP IdePix-Water Module v7.0.2, Operator Idepix.Meris v3.0
 - OLCI: SNAP IdePix-Sentinel3 Module v12.0.2, Operator idepix-olci-12.0.2.nbm
 - o The *IdePix* (Identification of pixel categories) neural network routine is applied for initial pixel identification as water, land, cloud/haze, cloud shadow or snow/ice. *IdePix* is called through SNAP using the GPT. Pixel identification masks are stored for later masking of invalid (non-water) pixels.
- Atmospheric correction.
 - o Tools:
 - Polymer v4.14

- Polymer according to Steinmetz et al. (2011), updated in Steinmetz (2016 and 2018), parameterised to use the Park and Ruddick (2005) bidirectional reflectance distribution function, and operating only on pixels identified as water by the *IdePix* module.
- Polymer is called using a function wrapper in Python.
- Optical water type classification.
 - Tools: Python implementation
 - Algorithm: the applied OWT is based on the work of Moore et al. (2001) and equivalent software developed for ESA CCI. The algorithm uses spectral means and covariance matrices defined in the UK NERC GloboLakes project (Spyrakos et al. 2018). In contrast to OWT mapping used in previous projects, we apply the spectral angle (Kruse et al. 1993) rather than Mahalanobis distance as metric for similarity between spectra.
 - The OWT classification is applied to each pixel to determine the similarity of the observed water-leaving reflectance spectrum to thirteen known OWTs. Two additional OWTs are used for identifying pixels contaminated by land adjacency (Jiang et al. 2023).
- Water constituent retrieval.
 - Algorithms: Water constituent retrieval algorithms tuned to each OWT in the GloboLakes project (Spyrakos et al. 2018, Neil et al. 2019) are mapped to each individual pixel according to the OWT with the three highest classification scores for that pixel. The algorithm results corresponding to those three OWTs are averaged using the membership score as weighting. The weighting factors are scaled between 0 and 1, where 1 is the highest OWT membership score and 0 is the first OWT score that is not included in averaging (i.e., the fourth). This procedure is used to derive maps of TSM, TUR, and Chl-a.
 - TSI is then calculated from Chl-a. The chlorophyll concentration is translated into 11 TSI classes following the categories in Table 1.
- Algorithm mapping and blending.
 - For each OWT a best-performing algorithm has been selected and tuned against the global in situ reference [Limnades](#) database, as part of the UK NERC GloboLakes project. For CGLOPS, the best performing algorithm corresponding to the OWT of a given pixel is mapped for turbidity, TSM and Chl-a as a precursor for trophic state.
- Floating cyanobacteria/macrophytes indicator.
 - Detection of surface blooms by applying the Maximum Peak Height (MPH) algorithm (Mathews and Odermatt 2015) adapted to use top-of-atmosphere (TOA) reflectance instead of Bottom-of-Rayleigh reflectance (BRR). The thresholds of the MPH were adapted to generate the same cyanobacteria surface bloom coverage from TOA signals compared to BRR, which reduces computation time. The algorithm differentiates between floating cyanobacteria and floating vegetation using spectral indices.

- Aggregation.

- Tool: SNAP LakeAggregate operator v1.0.
- LakeAggregate performs spatial and temporal aggregation of pixel values into cells ('bins') of a planetary grid over a lake. During the Lake Aggregation, invalid pixels are masked before averaging. Mean values (arithmetic averages) in each spatiotemporal bin are derived for the remaining observations. Invalid pixels are recognised based on L1b product flags (L1.INVALID), *IdPix* cloud, ice, haze, and land identification flags, and value limits for individual products (see the quality flag definitions for chl-a, TSM and turbidity in Table 2).
- Most representative spectrum selection: The algorithm has been developed by PML and BC for use in CGLOPS. For each pixel, water-leaving reflectance spectra are compared to the median spectrum using the root-mean-square (RMS) difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.
- 10-day aggregation is done on a per-lake basis after applying masks to select data for water pixels. The aggregated products which contain the reflectance, trophic state, turbidity, trophic state, TSM, Chl-a and cyanobacteria risk index variables are then mapped to a global grid at 0.0022° (nominally 300m) resolution. The global extent is UL: 180°W-90°N, BR: 180°E, 90°S.

- Quality flags:

- For the bio-geophysical variables, each pixel in the final products is an aggregation of multiple observations. The number of underlying observations will vary as some inputs are masked due to insufficient quality, contamination by cloud or ice and other factors. The *num_obs* variable gives the total number of input observations to a pixel before quality control. A *quality_flag* is provided which indicates what masking has removed input observations from the aggregated final output. Flags raised in this variable do not indicate a degradation in the quality of the final aggregated pixel, but a reduction in the number of input observations, which may reduce the representativeness of the result over the aggregation period. Table 2 shows the meaning of these quality flags.

The SNAP version used is 7.

Table 1: Trophic State Index and related chlorophyll concentration classes (according to Carlson (1977))

Trophic classification	TSI, CGLOPS values	Chl-a (µg/L) (upper limit)
Oligotrophic	0	0 - 0.04
	10	0.04 - 0.12
	20	0.12 - 0.34
	30	0.34 - 0.94

Trophic classification	TSI, CGLOPS values	Chl-a (µg/L) (upper limit)
Mesotrophic	40	0.94 - 2.6
	50	2.6 - 6.4
Eutrophic	60	6.4 - 20
	70	20 - 56
Hypereutrophic	80	56 - 154
	90	154 - 427
	100	427 - 1183

Table 2: Quality flag definitions

Flag name	Definition	Explanation
bright_pixels	$R_{w412} > 0.4$ or $R_{w865} > 0.4$	One or more input observations removed as anomalously bright
cloud	pixel_classif_flags.IDEPIX_CLOUD or pixel_classif_flags.IDEPIX_CLOUD_BUFFER or pixel_classif_flags.IDEPIX_CLOUD_SHADOW or pixel_classif_flags.IDEPIX_CLOUD_AMBIGUOUS or pixel_classif_flags.IDEPIX_CLOUD_SURFACE	One or more input observations removed as were flagged by <i>Idexpix</i> as being affected by cloud
snow_ice	pixel_classif_flags.SNOW_ICE	One or more input observations removed as flagged by <i>Idexpix</i> as snow or ice
I1_invalid	I1_flags.invalid	One or more input observations removed as instrument data missing or invalid, flagged in source data
Land	land_water_fraction < 100	One or more input observations removed as some percentage of

Flag name	Definition	Explanation
		the pixel is land
land_contaminated	(owt_14 + owt_15 > 1.59) and ((owt_dominant_class == 14) or (owt_dominant_class == 15))	One or more input observations likely contaminated by adjacency effect, but not removed (source variables for this expression not supplied)
atmospheric_correction_failure	Polymer bitmask: OUT_OF_BOUNDS	One or more input observations removed as Polymer bio-optical model reached its concentration limits
turbidity_invalid	turbidity > 100000 or turbidity < 0	One or more input observations removed as turbidity value likely unrealistic (suspected unmasked ice)
chl_a_invalid	Chl_a < 0 or Chl_a > 10000	One or more input observations removed as chl-a value likely unrealistic
tsm_invalid	TSM < 0 or TSM > 10000	One or more input observations removed as TSM value likely unrealistic
floating_cyanobacteria	floating_cyanobacteria > 0.4 and not (l1_invalid, cloud, land, land_contaminated)	Floating cyanobacteria likely in one or more input observations
floating_macrophytes_or_mixed_pixel	floating_macrophyte_probability (not supplied) > 0.5 and not (l1_invalid, cloud, land, land_contaminated)	One or more input observations likely contained floating macrophytes or was a mixed pixel

3.2.2 Basic underlying assumptions

The processing chain is subject to several sources of uncertainty including ambiguity in classification, algorithm calibration, and observation uncertainty. The validity of the resulting products is subject to validation and is ultimately based on a number of theoretical and practical assumptions on the performance of individual processing elements:

Assumptions addressed in product validation (see Quality Assessment Report):

- *IdePix* can differentiate adequately between pixels containing water and any of the following conditions: mixed land/water, cloud, cloud shadow, ice, snow, and haze.
- Polymer successfully retrieves the shape and amplitude of water-leaving reflectance.
- Tuning of reflectance algorithms for chlorophyll-a, suspended matter, and turbidity is adequate for each water type.
- The water-leaving reflectance spectra can be assigned a membership score for at least one known optical water type.
- The tuned reflectance algorithms for chlorophyll-a, suspended matter and turbidity yield valid results over the range of reflectance observed over water types to which they have been assigned.
- Selection of the most representative spectrum from observations obtained over 10 days uses statistical metrics that result in a consistent pattern for the whole water body.
- The translation of chlorophyll-a to Trophic State Index is adequate and consistent between water types.

Fundamental assumptions that are currently not independently addressed:

- The 13 optical water types defined in UK NERC GloboLakes sufficiently capture the natural optical diversity of inland waterbodies. More in situ observation data are needed to assess current gaps.

The most crucial assumption of the *Calimnos* processing chain is that optical water types which have been defined from a large set of in situ data from optically complex waters (lakes, reservoirs, lagoons, estuaries, and coastal areas) can be assigned successfully to each satellite observation (pixel) containing open water. In practice, the assignment to an optical water type can fail due to various reasons, the most prominent being failure to identify the pixel as water (or as influenced by cloud or ice) by *IdePix*, or failure in the atmospheric correction yielding a reflectance spectrum that is incompatible with any of the algorithms used downstream. Misclassification of a given image pixel will normally be associated with a low classification score giving rise to caution in using the resulting products for that pixel. Solutions for misclassification and improvement of atmospheric correction are continuously being explored.

3.3 LIMITATIONS OF THE PRODUCT

As in previous product versions, a systematic underestimation of water-leaving reflectance is apparent in the current version, particularly for highly turbid lakes, when compared to in situ measurements. This bias has been investigated, and the Polymer algorithm was improved to better

handle highly reflective lake water. An updated version of Polymer (v4.17) addressing this issue and subsequent algorithm calibration was completed within ESA Lakes_cci and is expected to feature in the CLMS Service under product v3.0. When marking a major version upgrade, reprocessing of archived satellite records is usually undertaken for the service at the same time.

Calimnos assigns the most appropriate algorithms for chl-a, TSM and turbidity based on the per-pixel similarity to known optical water types. In practice, atmospheric correction may have systematic errors for some water types, leading to low membership scores. This, in turn, implies that suboptimal reflectance algorithms for chlorophyll-a and suspended matter may be selected. Solutions for this issue are being explored. The inclusion of per-pixel uncertainty is one step in this direction. However, these uncertainty estimates propagate from global (per-OWT) algorithm performance and may not accurately reflect specific observation uncertainty. For example, observations near land under atmospheric haze are likely to be less reliable than observations in open bodies of water under clear atmospheric conditions.

If the pixel identification and, thus, the flagging of erroneous pixels does not work properly, e.g., at cloud borders, thin clouds or ice cover, particularly at the sub-pixel level, the water-leaving reflectance and subsequent water quality parameter retrieval may fail. Users interested in lake averages rather than maps are advised to mask a buffer of one or more pixels from shorelines, where such effects are most difficult to control and flag. An upgrade for the lake ice detection was introduced with version 1.4 (in production between June 2020 and October 2024) and the most recent update was introduced in v2.1 (current product version) that uses a newly trained neural network for scene classification that improves vertical artefacts in the products, while also improving detection of snow/ice covered lake water pixels. Version 1.3 products (before June 2020) are affected by less performant ice detection, although a static ice mask was applied to mask locations where ice is likely to occur. It is noted that *due to improved flagging of ice-covered lakes in v1.4 onwards (in comparison to v1.3), users should expect to observe a reduction in the number of pixels for which products are reported due to more robust quality control. This particularly affects high latitudes in winter and early spring.*

The spatial blending of different algorithms and/or the averaging of different, very inhomogeneous days under partial cloud cover can lead to visually inconsistent maps. Lakes may show patchy patterns where coverage on different days of observation was incomplete, and if optical conditions in the lake changed over this period. This is considered normal behaviour for the dekadal product.

The radiometric correction of OLCI on Sentinel 3A is implemented based on system vicarious calibration results. OLCI-B currently follows pre-flight calibrations while results from the tandem flights are pending. OLCI-B gains will be applied from LWQ product version 3.0.

3.4 RELATED AND PREVIOUS APPLICATIONS

The current version of the *Calimnos* NRT processing chain is 2.1 (Table 3), which differs from version 2.0 by an updated version of *IdePix*. Version 2.1 uses *idepix-olci-12.0.2.nbm* which is a newly trained Neural Network (NN) for scene classification, that includes O2A harmonised bands. The latest NN produces a more homogeneous classification and the Sentinel-3 OLCI “smile effect”

is significantly reduced. In addition, snow/ice cover in lakes is much better recognised. Example products are presented in the ATBD.

Older product versions

The previous version of the Calimnos NRT processing chain was 2.0. This version differs from the second last version 1.4 in several ways. Firstly, the atmospheric correction processor was updated from Polymer v4.12 to v4.14. The out-of-bounds flag from Polymer was applied. Additional variables included in v2.0 are chlorophyll-a, Total Suspended Matter, and the floating cyanobacteria index. Previously, TSM was calculated and converted to TUR, whereas TSM was included in v2.0 as a separate product while TUR is now calculated independently using the Nechad algorithm (Nechad et al. 2010, 2016). Per-pixel uncertainty variables are supplied for chlorophyll-a, TSM, and water-leaving reflectance.

For OLCI, Polymer v4.12 and v4.14 provide equivalent outputs. Therefore, reflectance and trophic state products found in Copernicus products v1.4.0 and v2.0 are fully compatible. Turbidity in v1.4.0 was calculated by applying a scaling factor to TSM, whereas in v2.0 a separate algorithm is used. Turbidity equivalent to v1.4.0 can be obtained in v2.0 by applying the following scaling factor to TSM: $\text{Turbidity} = \text{TSM} * 1.17$.

The second last version of the 300m LWQ product was v1.4, which differed from version 1.3 by an updated version of *IdePix* with improved lake ice detection. Products generated as v1.3 used a static, and, therefore, likely incomplete, ice mask in the period 2016-2019 (OLCI sensors), whereas ice masking for the MERIS sensor (2002-2012) performed nominally. These products are currently available for the reprocessed period 2002-2019, whereas v1.4 was introduced in June 2020. The period Jan-Jun 2020 was filled using v1.4.

The first release for CGLOPS used *Calimnos* v1.1.0 for reprocessing of the MERIS archive. *Calimnos* v1.0.4 was used to produce the first global adaptive water quality dataset in the UK NERC GloboLakes project to relate water quality changes to environmental change in 1000 lakes globally. Compared to GloboLakes processing, the temporal aggregation procedure was improved to select Lake Surface Reflectance as the statistically most representative reflectance spectrum for the 10-day period, and to aggregate TSI from chlorophyll-a concentration.

The selection of lakes has also evolved. Earlier product versions included an initial set of 1000 lakes, which covered more than 50% of the global surface area of inland water bodies (1,409,030 km², not counting the Caspian Sea which is also included in the service). Since version 1.3 of the processing chain, the set of lakes has been extended to 4,265 waterbodies to a total of 2,166,023 km². The included waterbodies spanning a surface size range from 0.11 to 379,446 km².

Planned future versions

Version 3.0 is anticipated, based on recent development of a new set of algorithms calibrated to use the next iteration of Polymer which is designed to give improved and more consistent results, particularly in very turbid water. The anticipated LWQ product v3.0 will include updated water-leaving reflectance variables, derived bio-geochemical products, and product uncertainty where

this can be determined. The various versions of past, present, and future products are summarized in Table 3.

Table 3: Summary of Lake Water Quality (LWQ) product versions. Polymer versions marked with * are equivalent for OLCI

LWQ Product version	IdePix version	Polymer version	Other changes from previous version
1.2	6.x		
1.3	7.0.4	4.12*	OWT matching uses spectral angle not Mahalanobis distance
1.4	7.0.5	4.12*	Improved ice masking in <i>IdePix</i>
2.0	7.0.5	4.14*	Cyanobacteria product Separate turbidity and TSM algorithms Chl-a supplied as well as trophic state Per-pixel uncertainty provided (Chl-a, TSM) Additional mask bands Polymer out-of-bounds masking applied
2.1	12.0.2	4.14*	Improved scene classification based on newly trained Neural Network available in major <i>IdePix</i> update
3.0	13.x	4.17 or higher	<i>Major Polymer (atmospheric correction) upgrade. All algorithms relying on atmospherically corrected reflectance retuned for Polymer 4.17 or higher</i>

4 PRODUCT DESCRIPTION

4.1 PRODUCT COLLECTION

The CLMS LWQ-300m product collection that is available to users is as follows:

Satellite sensor	Product version	Period covered
Envisat MERIS	LWQ-300m v1.3	May 2002-April 2012 (Archive)
Sentinel-3 OLCI	LWQ-300m v1.3	April 2016-December 2019
Sentinel-3 OLCI	LWQ-300m v1.4	January 2020-October 2024
Sentinel-3 OLCI	LWQ-300m v2.0	September 2024-October 2025
Sentinel-3 OLCI	LWQ-300m v2.1	November 2025-ongoing

4.2 FILE NAMING

For the Lake Water Products, the naming convention is as follows:

c_gls_<Acronym>_<YYYYMMDDHHmm>_<AREA>_<SENSOR>_<Version>.<EXTENSION>

where

- <Acronym> is the name of the product, which is **LWQ300** in the case of Lake Water Quality 300m.
- <YYYYMMDDHHmm> gives the temporal location of the file. YYYY, MM, DD denote the year, the month, the day of the product. The day is the first day of the dekad average. Because time is not relevant due to temporal aggregation, HHmm is always set to 0000.
- <AREA> gives the spatial coverage of the file. In our case, <AREA> is either **GLOBE** or the abbreviation of the continent (Table 4). Spatial subsets are at time of writing under discussion for LWQ products.

Table 4: Naming convention and bounding box for continental subsets

Short name Continent	Continent	Bounding Box
GLOBE	global	180°W – 180°E, 90°N – 90°S

- <SENSOR> gives the name of the sensor used to retrieve the product, either MERIS or OLCI.
- The <Version> is **1.3**. for MERIS/OLCI and 1.4, 2.0 or **2.1** for OLCI
- <EXTENSION> is indicating the file format, which is **.nc** for netCDF4 files.

Example:

c_gls_LWQ300_200504010000_GLOBE_OLCI_V2.1.nc

4.3 FILE FORMAT

The file format is NetCDF CF1.8 The format of each parameter (band) is provided in Table 5.

4.4 COLLECTION IDENTIFIER

The collection **identifier** for version 2.1 of the LWQ-300m product is:

clms_global_lwq_300m_v2_10daily-nrt_netcdf

The **DOI citation** of the LWQ-300m product v2 collection is:

<https://doi.org/10.2909/801137b8-9575-43ef-a073-140b663cc61c>

4.5 PRODUCT CONTENT

4.5.1 Data File

The bands that are included in the 300m products are listed in Table 5.

Table 5: Specification of the bands of the Lake Water Products

Band name	Data type	Description
num_obs	int	Number of observations (including non-water observations)
first_obs	float	Number of days from the start of the sensing period until the first cloud-free observation
last_obs	float	Number of days from the start of the sensing period until the last cloud-free observation
turbidity_mean	float	Mean turbidity in the observation period
chla_mean	float	Mean of chlorophyll a concentration in the observation period
chla_uncertainty	float	Mean of relative uncertainty of chlorophyll a concentration in the observation period
tsm_mean	float	Mean of total suspended matter in the observation period

Band name	Data type	Description
tsm_uncertainty	float	Mean of relative uncertainty of total suspended matter in the observation period
R _w 400_rep (OLCI)	float	Fully normalised water-leaving reflectance corresponding to the most representative spectrum for the observation period, at the waveband indicated in the band name. Band names indicated as 'OLCI' in the left column only occur with that sensor.
R _w 412_rep	float	
R _w 443_rep	float	
R _w 490_rep	float	
R _w 510_rep	float	
R _w 560_rep	float	
R _w 620_rep	float	
R _w 665_rep	float	
R _w 674_rep (OLCI)	float	
R _w 681_rep	float	
R _w 709_rep	float	
R _w 754_rep	float	
R _w 779_rep	float	
R _w 865_rep	float	
R _w 885_rep	float	
R _w 900_rep	float	
R _w 1020_rep (OLCI)	float	
R _w _relative_uncertainty	float	Relative uncertainty in fully normalised water-leaving reflectance at reflectance peak
R _w _relative_uncertainty_unbiased	float	Relative unbiased uncertainty in fully normalised water-leaving reflectance at reflectance peak
R _w DayNum_rep	float	Age, in days, after the start of the observation period, of the most representative water-leaving reflectance spectrum
trophic_state_index	float	Trophic state index (TSI) obtained from chlorophyll-a observations, averaged over the observation period. The TSI has a value of 0, 10, 20, 30, 40, 50, 60, 70, 80, 90, or 100 corresponding to the classes given in Carlson

Band name	Data type	Description
		(1977), where 0-40 are oligotrophic, 40-60 mesotrophic, 60-80 eutrophic, and 80-100 hypereutrophic.
floating_cyanobacteria	float	Floating cyanobacteria index, with a range of values from 0 to 1 indicating probability of cyanobacteria presence – the closer the value is to 1, the higher the probability
Lat	float	Latitude in decimal degrees
lon	float	Longitude in decimal degrees
time	float	Time in days since 1-1-1970
crs	string	Coordinate reference system information
quality_flags	int	Flag band showing why input observations to a pixel have been included. Flags are l1_invalid, atmospheric_correction_failure, cloud, land, land_contaminated, snow_ice, bright_pixels, turbidity_invalid, chla_invalid, tsm_invalid, floating_cyanobacteria, floating_macrophytes_or_mixed_pixel

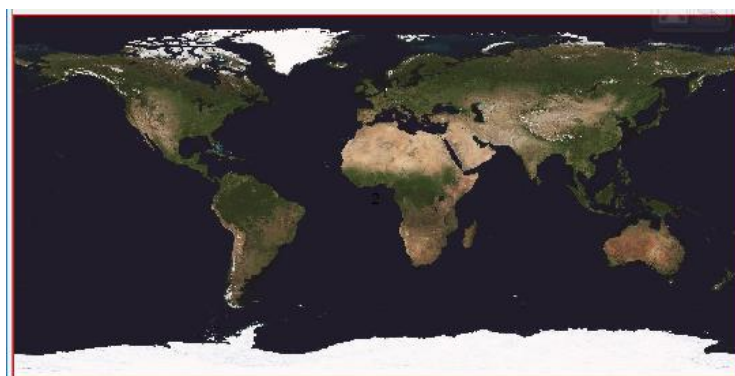
4.6 PRODUCT CHARACTERISTICS

4.6.1 Projection and Grid Information

Global, regional, or water-body specific files in NetCDF4 format, mapped to a global 0.0026° grid and including dimensions latitude, longitude (WGS84 projection), and time (days since 1 January 1970).

4.6.2 Spatial Information

The spatial extension of the global product covers the full globe, shown in the red rectangle in Figure 2.



Upper left latitude	89°59'45" N
Upper left longitude	179°59'45" W
Upper right latitude	89°59'45" N
Upper right longitude	179°59'45" E
Lower left latitude	89°59'45" S
Lower left longitude	179°59'45" W
Lower right latitude	89°59'45" S
Lower right longitude	179°59'45" E

Figure 2: Coverage of the global product (red rectangle) and bounding box coordinates

4.6.3 Temporal Information

The LWQ products are 10-day composites. The temporal information “YYYYMMDDHHmm” in the filename corresponds to the start date of the 10-daily period (called “dekads”). The 10D averages comprise the following days of the month: 1-10, 11-20, 21-31. As a consequence, the last “dekadal” period of the month varies in length between 8 and 11 observation days.

In addition to this (theoretical) observation time frame, the products contain the date of the (actual) first overpass contributing to the average and the (actual) last overpass. This information is stored in dedicated bands so that the observed period is known for each pixel (first_obs, last_obs). This information does not take into account whether a pixel is valid or not, but that a satellite observation was obtained.

The products from the NRT service are **available 3 days after the last acquisition** of the respective dekad period.

4.6.4 Data Policies

Any use of the Lake Water Quality products implies the obligation to include in any publication or communication using these products the following citation:

“The products were extracted from the land service of Copernicus, the Earth Observation program of the European Commission. The Calimnos production chain was initially developed within UK NERC GloboLakes and has been further developed with core collaboration between PML, University of Stirling, Brockmann Consult and HYGEOS in the Copernicus Land Monitoring Service, H2020-TAPAS, H2020-EOMORES, and ESA Lakes_cci.”

When using data or information from the service without modification:

“Copernicus Service information [Year].”

When using data or information from the service with modification:

“Contains modified Copernicus Service information [Year].”

Please send references of your publications using Copernicus Lake Water Quality products using our Service Desk <https://land.copernicus.eu/en/contact-service-helpdesk>.

For further details regarding the credits, copyright and legal ownership of the products, please refer to this Product User Manual and/or the relevant metadata, either in INSPIRE XML or inside the NetCDF data file.

4.6.5 Contacts

Accountable contact: European Commission Directorate – General Joint Research Centre

Email address: JRC-Copernicus-Land@ec.europa.eu

Scientific contact: BC and PML

Email address: lwq_info@brockmann-consult.de

Production and distribution contact: CLMS Helpdesk

<https://land.copernicus.eu/en/contact-service-helpdesk>

4.6.6 Sample products

The following figure provides examples of LWQ products for Lake Huron. It shows the Trophic State Index 10-day average (Figure 3 upper left), the turbidity 10-day average (Figure 3, upper right) and an RGB derived from the representative spectrum within the 10-day period (lower row).

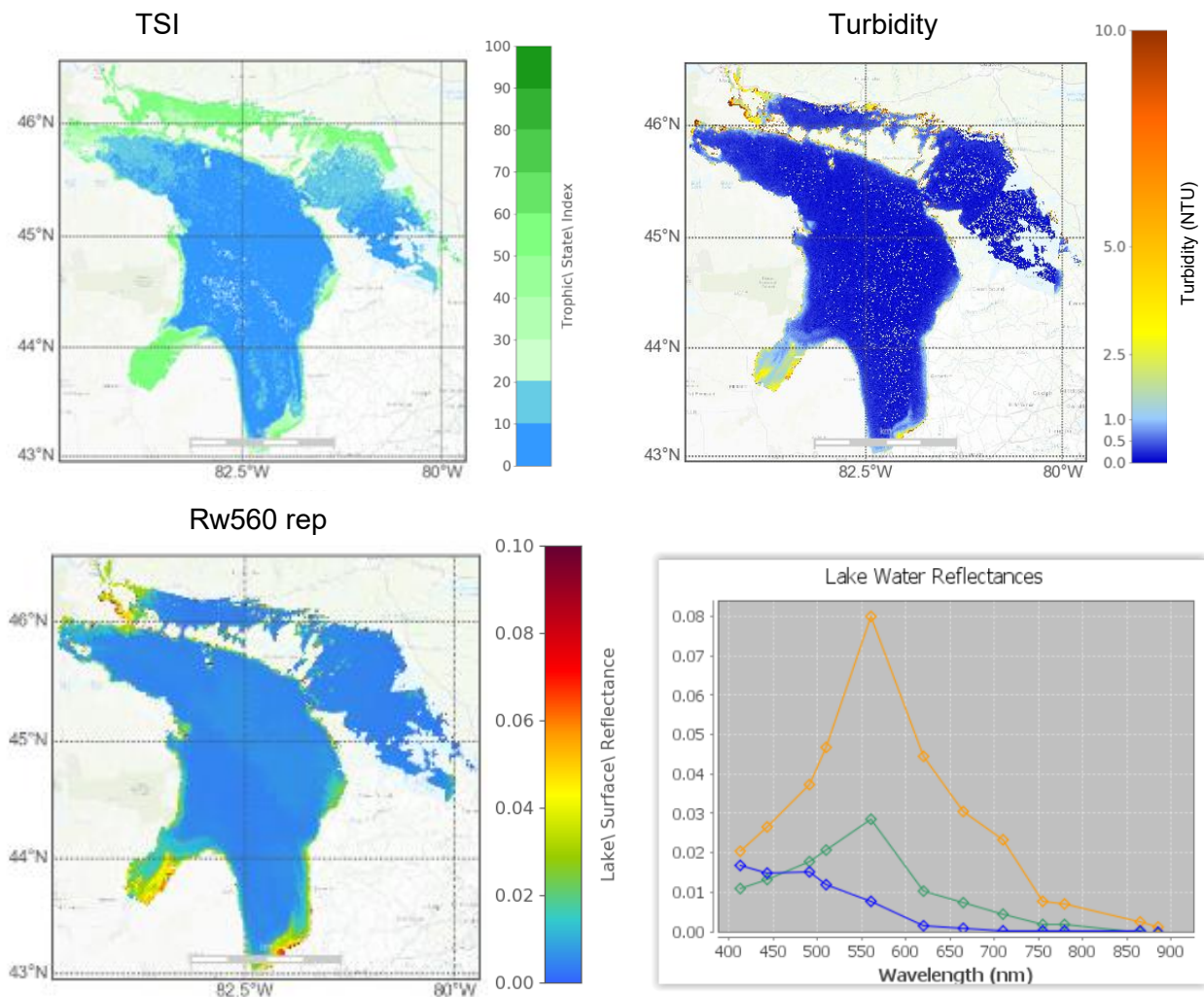


Figure 3: Lake Huron Trophic State Index (upper left) and turbidity (upper right), reflectance in band 560nm (lower left) dekadal product 2019-05-11 – 2019-05-20 and spectral reflectance plots showing all wavelengths for three positions in Lake Huron (lower right)

The dekadal products provide a good overview on the seasonal evolution of parameters by mapping them in sequenced maps. Figure 4 shows the dekads for the reflectance values at 550nm for Lake Garda. A clear seasonal trend is observed with increasing reflectance values in the green band towards the late spring and summer months. After August, the reflectance decreases again.

Figure 5 shows the dekads for turbidity in Lake Turkana for 2018. The distribution of suspended matter in the lake is clearly seen. Sediments are introduced to the lake via the Omo River in the

North and some smaller rivers in the west. The structures within the lake strongly depend on river runoff and, therefore, on raining and dry seasons.

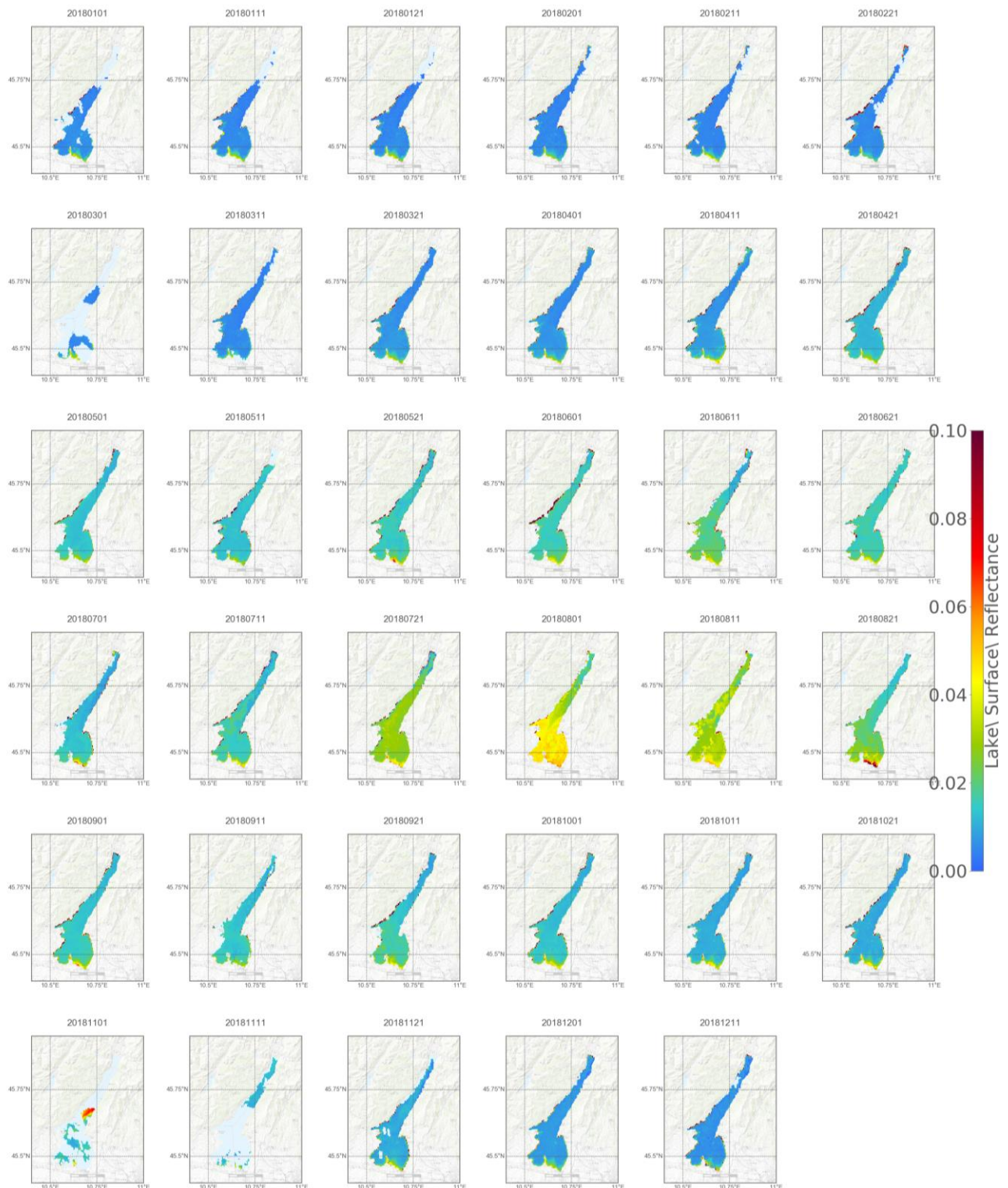


Figure 4: Maps showing the lake water-leaving reflectance at 560nm in Lake Garda (2018)

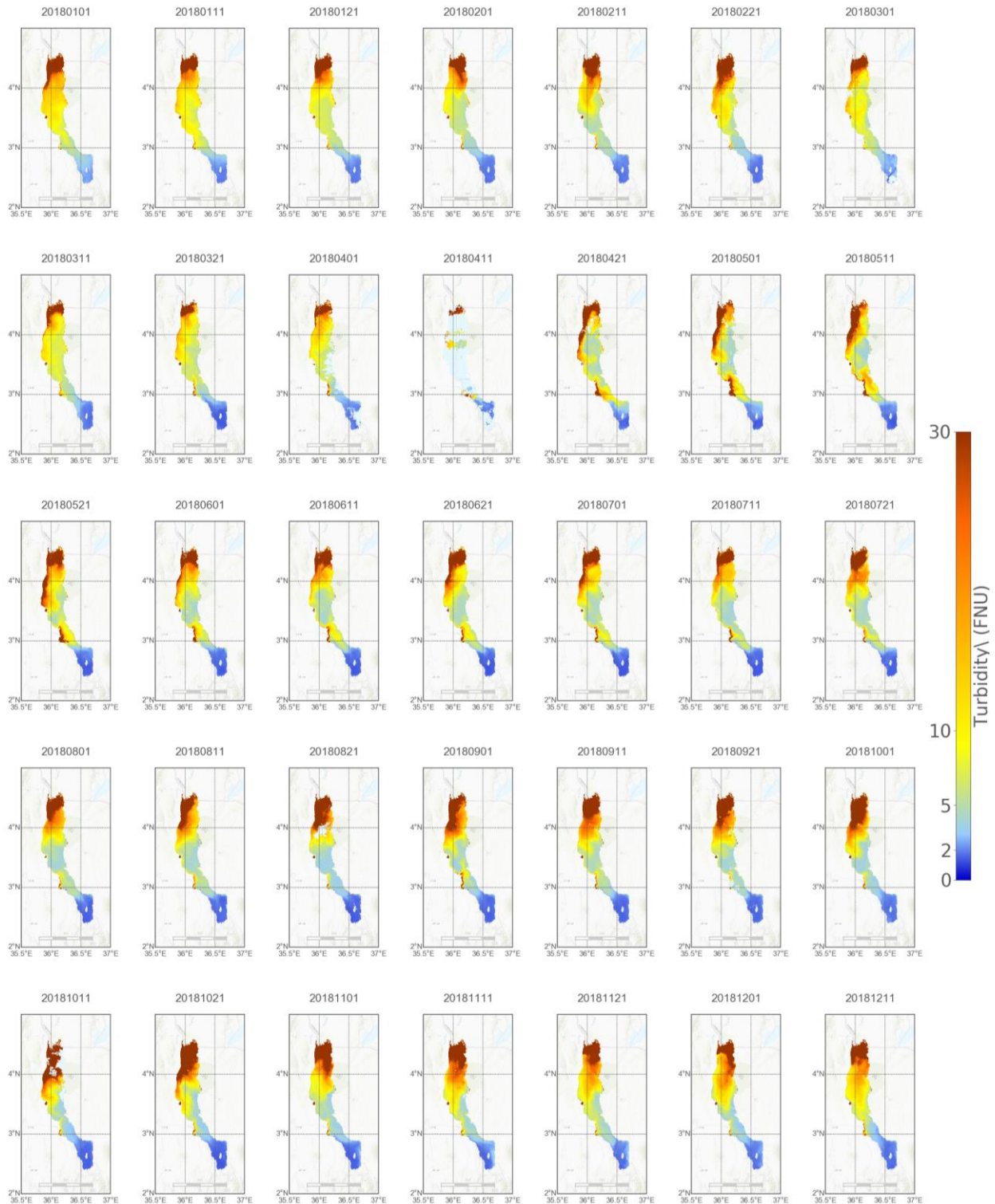


Figure 5: Turbidity dekad maps of Lake Turkana (2018)

5 VALIDATION

A detailed description of the validation and quality assessment is provided with the Quality Assessment Report (QAR) and a summary is provided herein. The products were tested against consistency (time series) and against in situ data, both for a selected set of lakes.

An earlier version of Polymer (v3.5) was validated against in situ reflectance data contained in the *Limnades* database during the UK NERC GloboLakes project (for MERIS). Polymer gave the best performance (unbiased error) in each waveband, although a consistent underestimation is apparent. This systematic error is cancelled out in the whole-chain calibration of TSM and chlorophyll-a retrieval. A correction is not attempted for the LSR because there are not enough in situ validation data available to inspect the performance per OWT. Further improvement of the Polymer atmospheric correction was based on results of further investigations, which showed that the mineral absorption model should not be universally applied, as was previously done. It should be noted that underestimation of reflectance has a minimal effect on water constituent retrieval algorithms that operate primarily on the shape of the reflectance spectrum. This is the case for turbid water chlorophyll-a algorithms and some TSM algorithms.

Meanwhile, Polymer has been further developed and the evolution between v3.5 and v4.14 (the version used to process MERIS and OLCI) can be summarized as follows: Versions 2.0 and 2.1 of the Lake Water Quality processing use Polymer v4.14. The previous LWQ product versions v1.3 and v1.4 used Polymer v4.12. As used in this processing chain there are no significant differences in the Polymer versions, therefore the reflectance data from LWQ product versions 1.3, 1.4, 2.0 and 2.1 are equivalent.

Time series comparison between in situ data and EO derived parameters enable to assess the behaviour of both measurement techniques over time. The focus is on the consistency of the time series on the one hand and on the comparability of the datasets on the other hand. The order of magnitude and seasonal patterns are investigated. The assessment of sample lakes shows that the products are consistent in time and mainly also in space. Seasonal patterns are as expected for the known and investigated lakes. Turbidity sometimes shows spatial patterns that are caused by temporal averaging of patchy (partly cloudy) products. The comparison between the products and in situ data shows the same magnitude, but only a few analyses could be performed.

Manual inspection of all products for more than 4000 water bodies is impossible and, in most cases requires local knowledge. The validation of the products is, and always will be, based on a small sample of well-studied areas. Users of these products are, therefore, advised to inspect the results for their area of interest before generating derivative products. This inspection could include, for example, histograms to identify outliers. Users are also advised to take into account the number of observations underlying the results. Where observations are sparse, having a small number of satellite overpasses to cover a large water body can lead to visual inconsistencies that do not reflect the state of the water body at any particular time – this is merely the nature of creating aggregated products.

Expert users are encouraged to take part in the calibration and validation of these products that is increasingly taking place at global scale. The spatiotemporal coverage and quality of the global lake water products can only be improved if the algorithms underlying these products can be accurately adjusted to waters of each optical type (and in some cases, new water types may need to be defined). We point interested users to the *Limnades* initiative (<https://www.globolakes.stir.ac.uk/limnades.html>), from where many of the presented validation results are derived.

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